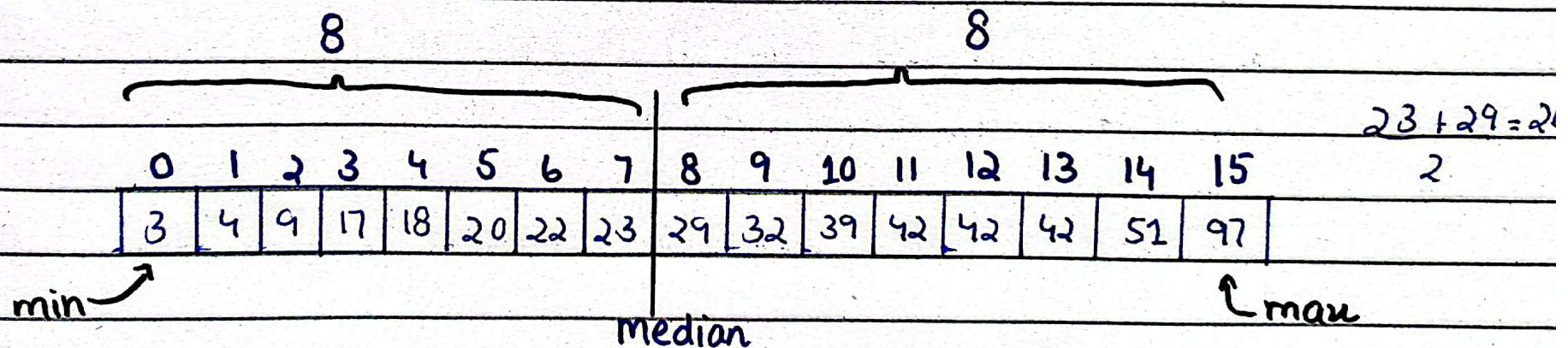
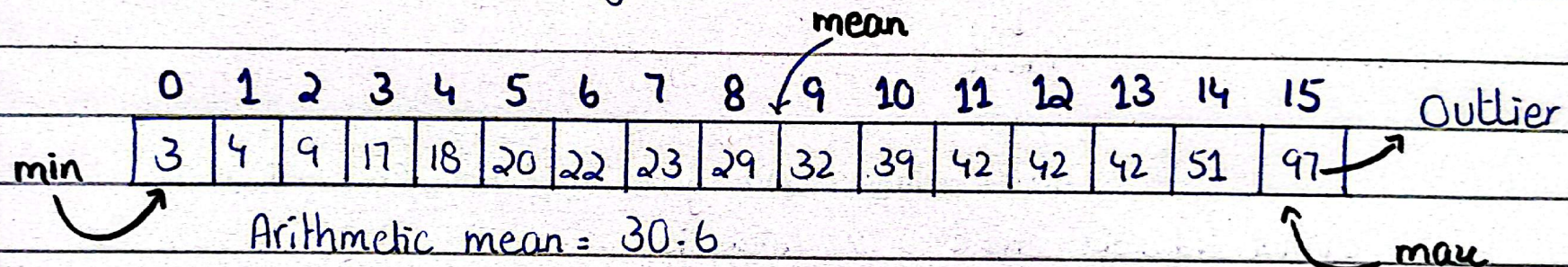


Lec 05:- Central tendency, Median, Quartiles

A specific person that can effect our whole average and make a big change in it we call it outlier.

how to get rid of that outlier.



median = 26

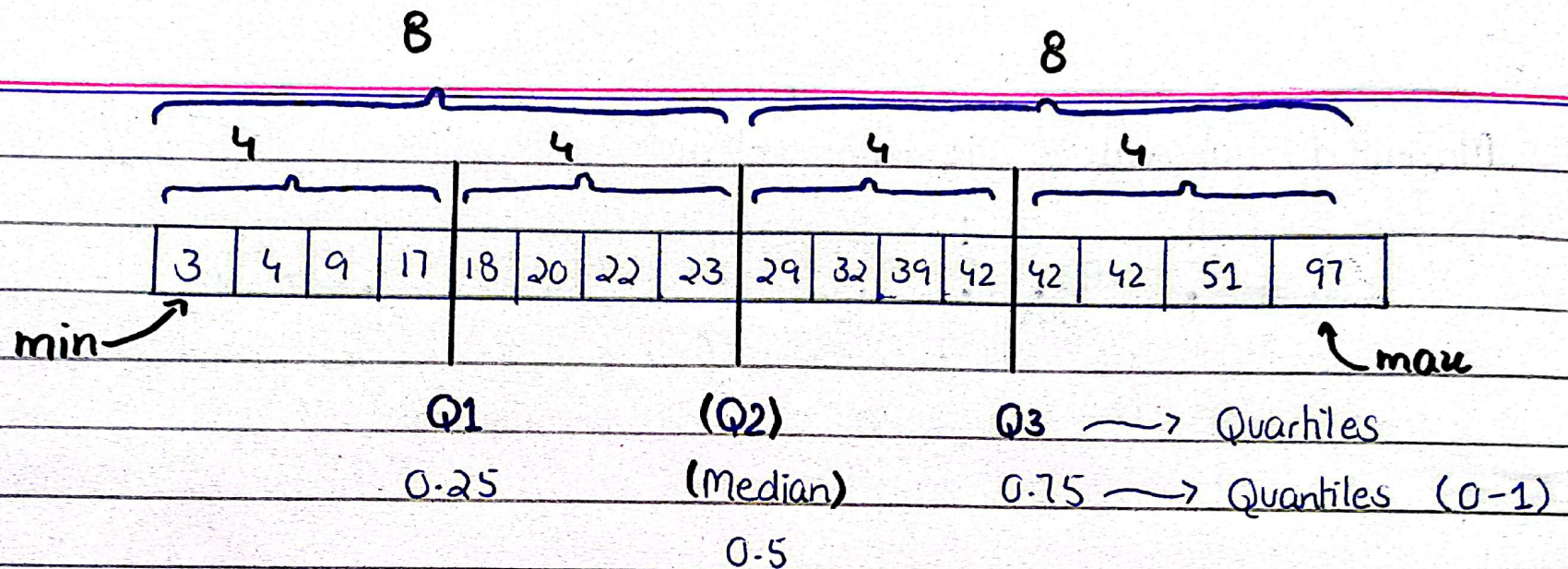
197 $\leadsto$ 26

35 $\leadsto$ 26

median doesnot effected by the outliar means that the
medians are "Robust to outliers"

↓
means
↓

Your outliers and your untorseen circumstances
cannot adversely effect your analysis.



Computing quantiles:-

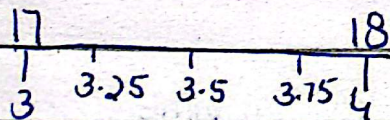
$$0.25 \leadsto \left\{ 0.25 (16 + 1) \right\}^{\text{th}} \text{ value}$$

$\downarrow$

4.25^{th} value

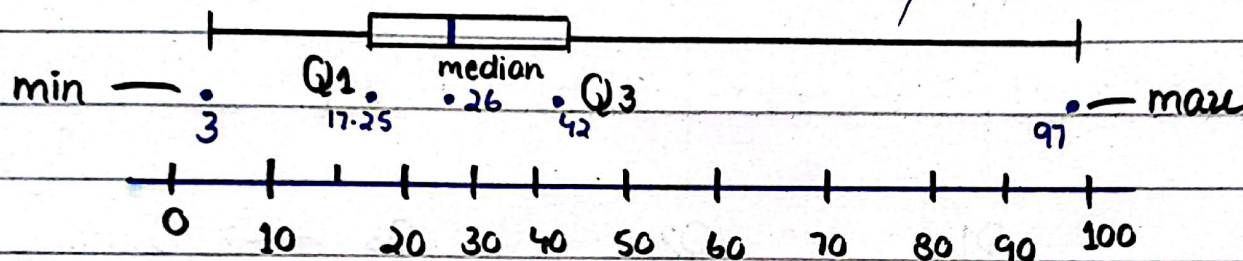
due to 0 base index (-1)

3.25^{th} value



$$\text{quantile } (0.25) = 17.25$$

Magnitude of values is not clear!



* we will see three types of average:-

- averages are measure of central tendency
- arithmetic mean which is sum of every thing divide by count of every thing. not robust to outlier. outlier effects our result.
- mode is most common value.
- median is robust to outlier its related is quantiles or quartiles if we divide our data in parts it tells in which part most people are.